Divisor concentration and primitive bases

Human Compression project: working theory for F008

This working note makes explicit the analytic and divisibility arguments used in the planar Mondrian density theorem of Lebowitz–Lockard and Tenenbaum [1]. The moment method is that of Hall and Tenenbaum [2]; it is not claimed here as a new method. We give the prime correlation argument, the passage from squarefree to arbitrary integers, and the passage from a Dirichlet series to an ordinary mean. The classical inputs retained below are Chebyshev's bound $\sum_{p\le z}\log p \ll z$, Mertens' bound $\sum_{p\le z}p^{-1}\le \log\log z+O(1)$, and elementary Euler summation for ζ. Constants may depend on the fixed parameters.

1 Moments of divisors in a short interval

Write

$$\Delta(n,u)=\#\{d \mid n : e^u<d\le e^{u+1}\},\quad \Delta(n)=\sup_u \Delta(n,u),\quad M_j(n)=\int_{\mathbb{R}}\Delta(n,u)^j\,du.$$

We use $\log_2 x=\log\log x$ and $\log_3 x=\log\log\log x$.

Lemma 1. *For $j\ge 1$, $M_1(n)=\tau(n)$, $M_j(n)\le \tau(n)^j$, and $\Delta(n)\le 2M_j(n)^{1/j}$. Also $\Delta(ab)\le \tau(b)\Delta(a)$ for arbitrary positive integers a,b. For $1<\sigma\le 2$,*

$$\sum_p \frac{\log p}{p^\sigma}<\frac{1}{\sigma-1}.$$

Proof. Each divisor contributes an interval of length one to the first integral. The bound $\Delta(n,u)\le\tau(n)$ gives the second assertion. If a unit interval contains D logarithms of divisors, integration over the interval of length two covering all their unit windows gives an integral at least D. Hölder's inequality gives $D\le 2^{1-1/j}M_j(n)^{1/j}$.

Every divisor of ab can be assigned uniquely a decomposition $d=vw$ with $v\mid a, w\mid b$: at each prime assign to v the smaller of its exponent in d and in a. For each fixed w, the possible v in a unit logarithmic interval contribute at most $\Delta(a)$.

For clarity, the last estimate is the elementary prime-series lemma of [2, Lemma 1]. Put $s=\sigma-1$. Euler summation gives

$$\zeta(\sigma)=\frac{1}{s}+\frac12+\frac{\sigma(\sigma+1)}{2}\int_1^\infty \frac{\{v\}(1-\{v\})}{v^{\sigma+2}}\,dv.$$

Thus $\zeta(\sigma)>s^{-1}+1/2$. Differentiating the integral, dropping the negative term and using $\{v\}(1-\{v\})\le 1/4$ gives $-\zeta'(\sigma)<s^{-2}+\sigma/(8(\sigma+1))$. For $0<s\le 1$ the quotient of this bound by $s^{-1}+1/2$ is smaller than s^{-1}. Finally the Euler product gives $\sum_p(\log p)p^{-\sigma}\le -\zeta'(\sigma)/\zeta(\sigma)$. □

Lemma 2 (Prime correlation)**.** *Fix an integer $r\ge 1$ and put $a=r+1$. For squarefree m, a prime $p\nmid m$, and an integer $j\ge 2$, write*

$$M_j(m^r p^r)=aM_j(m^r)+W_j(m,p).$$

Then $W_j(m,p)\ge 0$ and

$$\sum_{p\nmid m}\frac{\log p}{p}W_j(m,p)\ll_r a^j\tau(m^r)M_{j-1}(m^r).$$

Proof. Set $A_i(u) = \Delta(m^r, u - i\log p)$ for $0 \le i \le r$. The identity $\Delta(m^r p^r, u) = \sum_i A_i(u)$ shows that the pure terms in its jth power integrate to $aM_j(m^r)$. Every mixed monomial of degree j is bounded by $\sum_{i\neq h} A_i^{j-1} A_h$: choose a largest participating factor and one other participating factor. The sum of all multinomial coefficients is a^j.

For fixed $i \neq h$, change variables to $v = u - i\log p$. For each divisor $d \mid m^r$, the condition contributing to $\Delta(m^r, v + (i-h)\log p)$ restricts $\log p$ to an interval of length $1/|i-h| \le 1$. Chebyshev's bound and partial summation give a uniform upper bound for $\sum(\log p)/p$ on every such interval, including intervals meeting $p = 2$. Summing over d therefore bounds the prime sum of this mixed integral by $O(\tau(m^r)\int \Delta(m^r, v)^{j-1}\,dv)$. There are $a(a-1)$ ordered pairs, proving the assertion. □

Lemma 3 (Ordinary means)**.** *For fixed $r \ge 1, t > 0, y > 0$, let $f(n) = \Delta(n^r)^t y^{\omega(n)}$. Uniformly for $x \ge 3$,*

$$\sum_{n\le x} f(n) \ll_{r,t,y} \frac{x}{\log x} \sum_{n\le x} \frac{f(n)}{n}.$$

The same assertion holds for $f(n) = y^{\omega(n)}$.

Proof. Lemma 1 implies $f(mp^\nu) \le B_\nu f(m)$ with $B_\nu = \max(1, y)(r\nu + 1)^t$, whether or not $p \mid m$. Since $\log n = \sum_{p^\nu \mid n} \log p$, nonnegativity gives

$$\sum_{n\le x} f(n)\log n \le \sum_{m\le x} f(m) \sum_{p^\nu \le x/m} B_\nu \log p \ll x \sum_{m\le x} \frac{f(m)}{m}.$$

Indeed the $\nu = 1$ sum is $O(x/m)$ by Chebyshev. For $z = x/m$, the higher powers contribute

$$O(\sqrt{z}(\log z)^{t+1}) = O(z),$$

with the constant covering bounded z. For $n > \sqrt{x}$ we can divide by $\frac{1}{2}\log x$. For the remaining terms the elementary divisor bound gives $f(n) \ll n^{1/2}$, hence a total $O(x^{3/4})$, which is absorbed by $x/\log x$ since the harmonic sum is at least $f(1) = 1$. For the function $y^{\omega(n)}$, take $B_\nu = \max(1, y)$ in the same argument. □

Theorem 4 (Weighted concentration)**.** *Fix an integer $r \ge 1$, real $t > 0, y > 0$, and $\kappa > \sqrt{2t}$. Put*

$$\gamma_r(y, t) = y - 1 + \big((r+1)^t y - y - t\big)^+.$$

Then, as $x \to \infty$,

$$\sum_{n\le x} \Delta(n^r)^t y^{\omega(n)} \ll x(\log x)^{\gamma_r(y,t)} \exp\{\kappa\sqrt{\log_2 x \log_3 x}\}.$$

Proof. Put $a = r + 1$. First suppose $y \ge t/(a^t - 1)$, and choose an integer $j \ge \max(2, t)$. Define the absolutely convergent series

$$L(\sigma) = \sum_{m \text{ squarefree}} \frac{M_j(m^r)^{t/j} y^{\omega(m)}}{m^\sigma}, \qquad 1 < \sigma \le 2.$$

Absolute convergence, also after differentiation, follows from $M_j(m^r) \le a^{j\omega(m)}$ and the corresponding Euler product. Differentiation singles out one prime. Concavity of $v^{t/j}$ separates the pure and mixed terms of Lemma 2. Hölder in the prime sum, with weights $(\log p)/p^\sigma$, and Lemma 1 give

$$-L'(\sigma) \le \frac{ya^{t/j}L(\sigma)}{\sigma - 1} + \frac{yC_r^{t/j} a^t}{(\sigma-1)^{1-t/j}} \sum_m{}^{*} \frac{y^{\omega(m)}[\tau(m^r) M_{j-1}(m^r)]^{t/j}}{m^\sigma}.$$

Here and below a star means squarefree. Interpolation between M_1 and M_j gives

$$\tau(m^r)M_{j-1}(m^r) \le a^{j\omega(m)/(j-1)} M_j(m^r)^{(j-2)/(j-1)}.$$

For $j = 2$ this is the equality $\tau M_1 = \tau^2$. Another application of Hölder, now to the m sum, therefore yields

$$-L'(\sigma) \le \frac{bL(\sigma)}{s} + \frac{BL(\sigma)^{(j-2)/(j-1)}}{s^{1+\alpha/(j-1)}}, \quad s = \sigma - 1,$$

where

$$b = ya^{t/j}, \qquad \alpha = a^t y - t + t/j, \qquad B = yC_r^{t/j} a^t 2^{a^t y/(j-1)}.$$

We used $\sum_m^* (a^t y)^{\omega(m)} m^{-\sigma} \le \zeta(\sigma)^{a^t y} \le (2/s)^{a^t y}$. The coefficient B is bounded independently of j.

The difference $\alpha - b$ needs a positive lower bound. At the smallest allowed y it equals

$$\frac{t}{j} - \frac{t(a^{t/j} - 1)}{a^t - 1}.$$

Since $j(a^{t/j} - 1) \le 2(a^{t/2} - 1)$ for $j \ge 2$, convexity gives

$$\alpha - b \ge \frac{t}{j} \frac{a^{t/2} - 1}{a^{t/2} + 1} = \frac{c_t}{j} > 0.$$

Increasing y only increases this difference. This replaces the overly strong displayed bound $\alpha - b \ge t/j$ in the source proof.

Set $Y = L^{1/(j-1)}$. The preceding differential inequality is linear in Y. Multiplication by $s^{b/(j-1)}$ and integration from s to one give

$$Y(1+s) \le \left(Y(2) + \frac{B}{\alpha - b} \right) s^{-\alpha/(j-1)}.$$

Since $L(2) \le \zeta(2)^{a^t y} < 2^{a^t y}$, it follows that

$$L(1+s) \le (Cj)^j s^{-a^t y + t - t/j}.$$

We now remove the squarefree restriction. Write $n = uv$, where u is the product of the primes appearing to exponent one in n and v contains the remaining prime powers. Then u is squarefree, v is powerful, and $(u, v) = 1$. Lemma 1 gives $\Delta(n^r) \le \tau(v^r)\Delta(u^r)$. The series

$$\sum_{v \text{ powerful}} \frac{\tau(v^r)^t y^{\omega(v)}}{v^\sigma} = \prod_p \left(1 + y \sum_{\nu \ge 2} \frac{(r\nu + 1)^t}{p^{\nu\sigma}} \right)$$

is bounded uniformly for $\sigma \ge 1$. Thus the full Dirichlet series of $f(n) = \Delta(n^r)^t y^{\omega(n)}$ is at most a fixed constant times $L(\sigma)$, using $\Delta(u^r)^t \le 2^t M_j(u^r)^{t/j}$. Lemma 3, with $\sigma = 1 + 1/\log x$ and $n^{-1} \le en^{-\sigma}$ for $n \le x$, now gives

$$\sum_{n \le x} f(n) \ll x(\log x)^{a^t y - t - 1} \exp\{(t/j)\log_2 x + j \log(Cj)\}.$$

For sufficiently large x, take

$$j = \lfloor \sqrt{2t \log_2 x / \log_3 x} \rfloor.$$

The last exponent is then $(\sqrt{2t} + o(1))\sqrt{\log_2 x \log_3 x}$.

Finally suppose $0 < y < t/(a^t - 1)$. The function $T \mapsto T/(a^T - 1)$ is continuous and strictly decreasing to zero. Hence there is a $T > t$ with $y = T/(a^T - 1)$. Apply the case just proved with T and then Hölder to

$$\sum_{n \le x} \Delta(n^r)^t y^{\omega(n)} \le \left(\sum_{n \le x} \Delta(n^r)^T y^{\omega(n)} \right)^{t/T} \left(\sum_{n \le x} y^{\omega(n)} \right)^{1-t/T}.$$

The final mean is $O(x(\log x)^{y-1})$: its Euler product is at most $C_y \zeta(\sigma)^y$, and Lemma 3 applies. At the chosen threshold both logarithmic powers are $y - 1$. The exponential constant can be chosen arbitrarily close to $t\sqrt{2/T} < \sqrt{2t}$. This proves the remaining case. □

2 Exceptional divisors and sets of multiples

For a set S of positive integers, let $\mathcal{M}(S)$ be the set of integers divisible by at least one member of S.

Lemma 5 (Excess prime factors)**.** *Fix $\varepsilon > 0$ and let*

$$E = \{b > 1 : \Omega(b) > (1 + \varepsilon) \log_2 b\}.$$

There is $\eta > 0$ such that the upper density of $\mathcal{M}(E \cap [N, \infty))$ is $O((\log N)^{-\eta})$.

Proof. Choose $1 < z < 2$ sufficiently close to one that $\eta = (1 + \varepsilon) \log z - (z - 1) > 0$. Let n_Y denote the part of n supported on primes at most Y. For every $X, Y \ge 2$, expansion as a divisor sum gives

$$\frac{1}{X} \sum_{n \le X} z^{\Omega(n_Y)} \le \prod_{p \le Y} \frac{1 - p^{-1}}{1 - zp^{-1}} \ll_z (\log Y)^{z-1}.$$

Indeed the divisor coefficients at p^ν are $(z - 1)z^{\nu-1} \ge 0$; replace each $\lfloor X/d \rfloor / X$ by $1/d$. Taking logarithms of the product and using Mertens' bound proves the last inequality; the terms of degree at least two sum absolutely.

If $e^{e^j} \le b < e^{e^{j+1}}$, $b \mid n$ and $b \in E$, then $\Omega(n_{e^{e^{j+1}}}) > (1 + \varepsilon)j$. Markov's inequality bounds the proportion of such $n \le X$ by $O(e^{-\eta j})$, uniformly in X and j. Summing over $j \ge \lfloor \log_2 N \rfloor$ proves the assertion, with a constant covering the first incomplete interval. □

Lemma 6 (Density below one)**.** *Suppose $S \subset \mathbb{N} \setminus \{1\}$ and the upper density of $\mathcal{M}(S \cap [N, \infty))$ tends to zero. Then $\mathcal{M}(S)$ has an asymptotic density strictly smaller than one. In particular this holds if $\sum_{b \in S} b^{-1} < \infty$.*

Proof. The multiples of a finite set have a periodic density. Their difference from $\mathcal{M}(S)$ lies in a tail whose upper density tends to zero. Their densities consequently converge, and the upper and lower densities of $\mathcal{M}(S)$ coincide. The same argument applies to every tail of S.

We recall the elementary finite form of Behrend's inequality. For finite sets A, B, the densities of their complements satisfy

$$d(\mathcal{M}(A \cup B)^c) \ge d(\mathcal{M}(A)^c) d(\mathcal{M}(B)^c).$$

To see this, sample an integer uniformly modulo the least common multiple of their entries. Its truncated prime valuations are independent. Each event of avoiding a prescribed collection of divisors is decreasing in those valuations. Two decreasing functions of one variable have nonnegative covariance, by expanding $\frac{1}{2}\mathbb{E}[(f(X) - f(X'))(g(X) - g(X'))] \ge 0$. Induction on the independent coordinates proves the same assertion for the product measure, giving the inequality.

Choose N so that the density of $\mathcal{M}(S \cap [N, \infty))$ is less than $1/2$. The finite set $A = S \cap [1, N)$ has complement of positive density: the residue class one modulo its least common multiple avoids every entry. Apply the finite inequality to A and larger finite parts of the tail, then pass to their densities. The complement of $\mathcal{M}(S)$ has density at least $d(\mathcal{M}(A)^c)/2 > 0$. Finally the upper density of a tail is at most its reciprocal sum. □

Boundary of this note. Theorem 4 recovers source Theorem 4.1 when $r = 2, t \geq 1$, and its unweighted-power specialization for general r. The proof also covers $0 < t < 1$: it only uses $0 < t/j \leq 1$, and no step requires $t \geq 1$. This additional range still awaits distinct review. It does not establish the source's full multiplicative-weight remark or its proposed improved exponential error. The three-dimensional geometric application remains separate. This working note has not received distinct review and adds no completed-original, conjecture or formal-verification credit.

References

[1] N. Lebowitz–Lockard and G. Tenenbaum, *On solutions to the Mondrian Puzzle*, retained thirteen-page preprint dated 28 August 2026.

[2] R. R. Hall and G. Tenenbaum, *The average orders of Hooley's Δ_r-functions, II*, Compositio Math. **60** (1986), 163–186.